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$$1 - i$$

$$r = |z| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$\tan(\theta) = \frac{1}{-1} = -1$$

$$\theta = \text{Bgtan}(-1) = \frac{-\pi}{4}$$

$$z = r \cdot \cos(\alpha) + i \cdot r \cdot \sin(\alpha)$$

$$= \sqrt{2} \cdot \cos\left(\frac{-\pi}{4}\right) + i \cdot \sqrt{2} \cdot \sin\left(\frac{-\pi}{4}\right)$$

$$= \sqrt{2} \cdot \text{cis}\left(\frac{-\pi}{4}\right)$$

References